

SERAPAT RATANAPACHAI

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EDUCATION

King Mongkut's Institute of Technology Ladkrabang

August 2022 – April 2026

Bachelor of Science in Information Technology

- GPA 3.16 / 4.00
- Relevant Courses: Software Development, System Analysis, System Design

TECHNICAL SKILLS & TECHNOLOGIES

- Languages: Python, Go, Java (OOP), JavaScript, TypeScript, Dart, C, C++
- Frameworks & Libraries: Next.js, React.js, FastAPI, Flutter, Django, Agno, Gin, Fiber
- Databases & Vector Search: PostgreSQL, MySQL, PGVector, SQLite, SQL Server, Supabase
- AI & Machine Learning: Retrieval-Augmented Generation (RAG), Agentic RAG, LLM Integration
- Tools & Infrastructure: Git, Docker, Postman, AWS S3

WORK EXPERIENCE

Software Developer Intern, Datapro Computer Systems Co., Ltd

April- June 2025

- Developed the end-to-end Incident Submodule (both Frontend and Backend) using Next.js, and Golang (Gin). SQL Server to modernize core workflows for a legacy TPA insurance platform, improving user data-handling efficiency.
- Designed and validated RESTful APIs and interactive API Documentation using Swagger, ensuring strict data validation
- Conducted an internal technical workshop for the engineering team on Golang (Gin Framework) principles, and Validator V10

PROJECTS

AI Chatbot Generator Platform [Senior Project]

Technologies used: Next.js, FastAPI, Agno, PGVector, PostgreSQL, AWS S3, Docker

- Developed a platform for AI chatbot creation and knowledge base management, enabling seamless access via iframes or shareable URLs, thereby solving the high technical barriers of traditional RAG deployments
- Co-developed an Agentic RAG pipeline to streamline data integration and retrieval, enabling 3 specialized agents (Document, PostgreSQL, API) to dynamically query databases and external data.
- Co-developed in integrating multi-LLM orchestration (Gemini, GPT, DeepSeek) and system architecture, applying prompt engineering to ensure accurate and grounded AI responses.

Virtual Asset Trading Platform

Technologies used: Next.js, Prisma, Supabase, Clerk, Tailwind CSS, shadcn/ui

- Developed a cryptocurrency paper trading platform that allowed users to simulate trading with virtual funds in a risk-free environment.
- Integrated CoinGecko API for real-time cryptocurrency market data and Stripe API for virtual fund top-ups, enabling a realistic trading experience.
- Implemented Google authentication using Clerk, along with portfolio management, transaction tracking, profit/loss calculation, and personalized watchlists.

Expert Connect

Technologies used: Next.js, Go, PostgreSQL, Cloud AWS Service

- Engineered a role-based task-matching platform utilizing Golang and PostgreSQL, enabling Askers to post specific queries with budgets and Experts to provide solutions.
- Developed an end-to-end task management workflow utilizing Next.js and Go to handle job requirements and expert business logic, seamlessly integrating AWS S3 for secure evidence submissions.
- Collaborated within a Multi-AZ AWS Cloud-native environment, ensuring the seamless integration of developed features with automated CI/CD pipelines via GitHub Actions.

Music Discovery & Social Review App

Technologies used: Flutter, FastAPI, OAuth 2.0, PostgreSQL, Spotify API, Deezer API

- Architected and developed a full-stack, cross-platform mobile application (iOS/Android) as the sole developer, engineering a multidimensional music review system (Beat, Lyric, Mood) with a dynamic, color-based emotion
- Engineered a scalable FastAPI backend with PostgreSQL and SQLAlchemy ORM, designing efficient relational database schemas to seamlessly manage secure user authentication.
- Synchronized Spotify and Deezer APIs to stream 30-second audio previews and fetch real-time music metadata, implementing a smart search function and strict backend data validation via Pydantic.

ADDITIONAL INFORMATION

- **Certifications:** Huawei Cloud Certified Developer Associate (HCCDA) – Tech Essentials Huawei Cloud